1.2_WithinPlot_LASSOofTruth

2025-06-20

The Goal of this document is to use LASSO regression to tell me which variable I can remove from my GLMM at the within-plot level of this project. I will be looking to cross-validate it as well. Then I can use that model and do my negligible effects magic as well. Aspect is the tricky one (yet again) but by doing a 'group lasso' with dummy variables, it should work out!

I may also try to explore Dr. Gilbert's idea of removing species or group of species and the simulating those results. Perhaps some kind of randomization test could also prove useful.

LASSO-ing

Okay, let's get the Lasso working.

All Plots

What I figured out is that there are two routes. I only have soil data for two sites but that cuts my sample size from 107 to 63. So there is a trade off between all the plots and all the predictor variables. I decided to pursue both routes to see if it told me anything. Spoiler, I think I'll just report both.

Data Wrangling

The first thing I have to do is get the data looking nice. I separate out the response variable. Then I select out the predictor variables I have data for every plot on. This has to be a matrix. I've set up Aspect as a dummy variable for each and every level. Any weirdness will be solved by grouping my predictors. That's what the 'AspectGroup' does. It will lump all my aspect dummy variables together and leave everyone else alone. The Lasso then will decide to include all or none of the Aspect Dummy variables.

```
head(Comb_Dis)
```

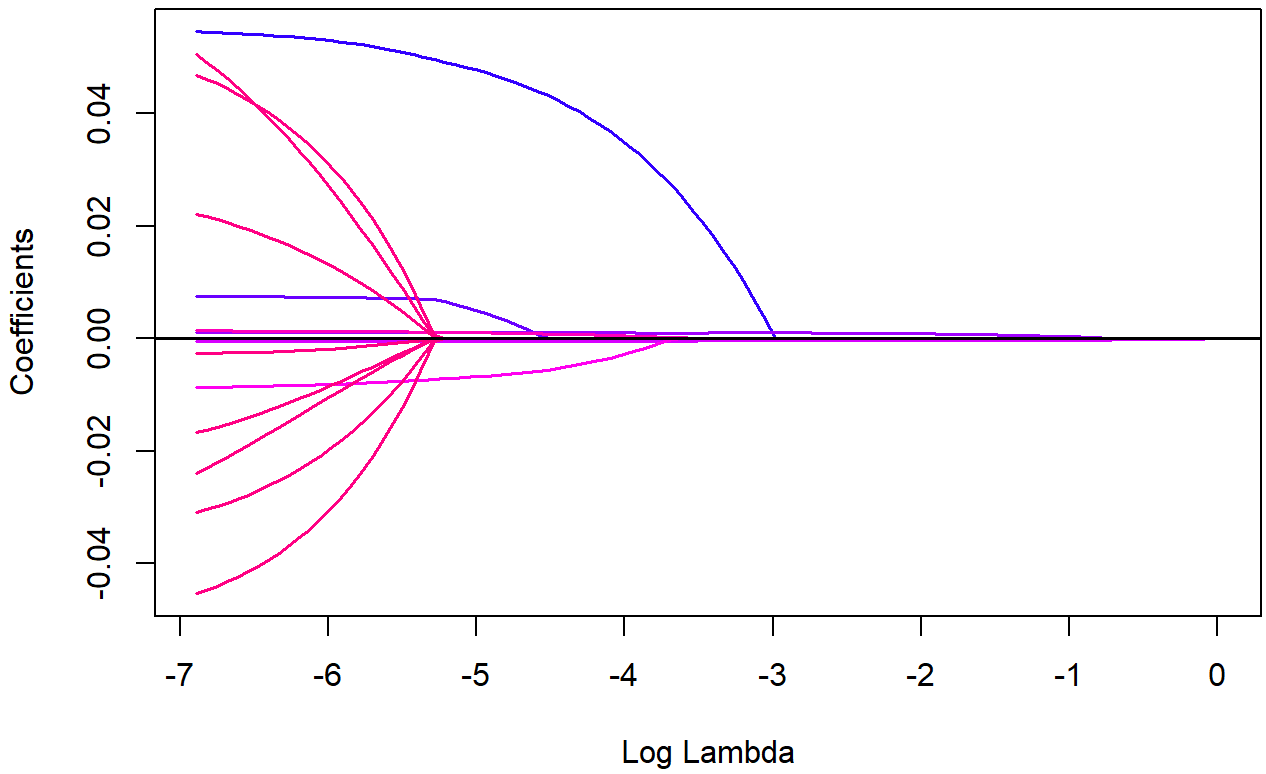
```
##   Plot Dissimilarity        X         Y          Z Site    SoilTemp  SoilMoist
## 1    1     0.4024599 27.99054 15.291673 0.09103439 TeMi -0.27407649  1.5065314
## 2    2     0.0000000 34.03762  7.180370 0.00000000 TeMi  0.37848659  1.1790246
## 3    3     0.4196000 46.89969  6.040091 3.30817681 TeMi  0.37848659  1.1790246
## 4    4     0.5428571 53.33120 14.105131 4.49339646 TeMi -0.01305126 -0.4585096
## 5    5     0.8193920 44.40650 30.406395 4.34845167 TeMi -0.14356388  0.8515178
## 6    6     0.0000000 35.08873 20.957058 4.65637884 TeMi  0.63951182  0.1965041
##   Slope Aspect     Date Collectors dry_weight   pH cond_us.cm salinity_ppt
## 1 15.84      S 8/8/2022  BJR + ARM      20.00 9.30     1461.0        0.576
## 2  2.71      N 8/8/2022  BJR + ARM      20.02 9.42      941.0        0.287
## 3  1.30      N 8/8/2022  BJR + ARM      20.03 9.12      883.4        0.319
## 4  4.75      S 8/8/2022  BJR + ARM      20.00 9.21      614.8        0.231
## 5  7.71     SE 8/8/2022  BJR + ARM      20.02 8.99      524.9        0.192
## 6 10.09      E 8/8/2022  BJR + ARM      20.02 9.28      740.0        0.262
##                                                     Notes
## 1 highlighted: mixed on 1/27, refrigerated, read on 2/1
## 2                                                    <NA>
## 3                                                    <NA>
## 4                                                    <NA>
## 5                                                    <NA>
## 6                                                    <NA>
```

```
With_Las <- subset(Comb_Dis, Dissimilarity >0.0) #this just rid of plots that were all sand

Las.y <- With_Las[,2]
DumAspect<-as.matrix(dummy_cols(With_Las, select_columns = "Aspect")[,18:25])
Las.x1<- With_Las %>% select(c( "SoilTemp", "SoilMoist","X", "Y", "Z", "Slope"))

AspectGroup <- c((1:dim(Las.x1)[2]), rep(dim(Las.x1)[2]+1, 8))


Las.x<- Las.x1%>%
  cbind(DumAspect) %>%
  as.matrix()
```

Actually run the Lasso

Okay, here I'm faced with yet another fork in the road. There is a standard lasso first which gives the coeffecients along a range of Lambdas. That's good but, then there is the cross validating lasso that used k-fold cross-validation that helps with over fitting. I think that's the move but then it leaves us with another fork! Do we want the minimum lambda (which is MSE) or a standard error away which one source said can help avoid overfitting. In this case, I don't think I'm stressed about overfitting so we are minimizing error (also, uh, no coeffecients ever are non-zero in the other one soooooo year)

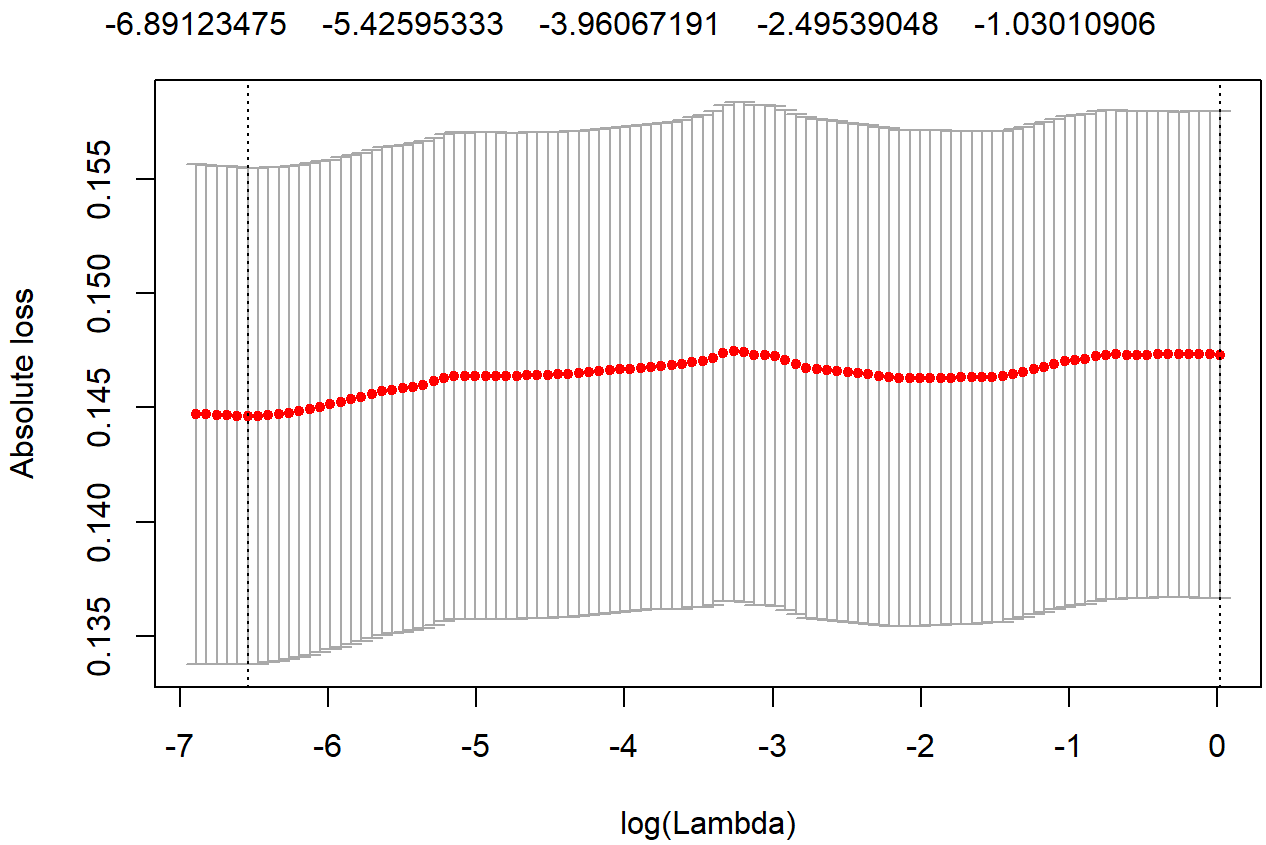
```
With.Lass<-gglasso(x = Las.x, y = Las.y, group = AspectGroup, loss = "ls")
plot(With.Lass)
```




```
coef(With.Lass)[,95:100]
```

```
##                       s94           s95           s96           s97
## (Intercept)  0.4623563860  0.4624386186  0.4624986496  0.4625391734
## SoilTemp     0.0541986493  0.0542932908  0.0543793871  0.0544577197
## SoilMoist    0.0074891926  0.0075062538  0.0075228307  0.0075389291
## X            0.0011794055  0.0011822003  0.0011850024  0.0011877933
## Y           -0.0005223101 -0.0005248664 -0.0005272439 -0.0005294546
## Z           -0.0085190466 -0.0085536308 -0.0085854496 -0.0086146793
## Slope        0.0013672451  0.0013744241  0.0013812422  0.0013877106
## Aspect_E    -0.0140461170 -0.0146403253 -0.0152061176 -0.0157436431
## Aspect_N    -0.0024582480 -0.0024936304 -0.0025224023 -0.0025451995
## Aspect_NE   -0.0278372971 -0.0285685901 -0.0292459579 -0.0298730720
## Aspect_NW   -0.0190157892 -0.0200624513 -0.0210895579 -0.0220947796
## Aspect_S     0.0424722720  0.0434998220  0.0444531494  0.0453379816
## Aspect_SE    0.0429493577  0.0446290112  0.0462302615  0.0477544857
## Aspect_SW    0.0194214547  0.0200417190  0.0206272081  0.0211796106
## Aspect_W    -0.0414852144 -0.0424050792 -0.0432460482 -0.0440147316
##                       s98           s99
## (Intercept)  0.4625627069  0.4625715312
## SoilTemp     0.0545290025  0.0545938871
## SoilMoist    0.0075545482  0.0075696859
## X            0.0011905570  0.0011932798
## Y           -0.0005315098 -0.0005334203
## Z           -0.0086414948 -0.0086660648
## Slope        0.0013938408  0.0013996449
## Aspect_E    -0.0162532735 -0.0167355409
## Aspect_N    -0.0025626529 -0.0025753481
## Aspect_NE   -0.0304534380 -0.0309903599
## Aspect_NW   -0.0230760444 -0.0240315240
## Aspect_S     0.0461595475  0.0469226556
## Aspect_SE    0.0492033275  0.0505786865
## Aspect_SW    0.0217005354  0.0221915481
## Aspect_W    -0.0447172713 -0.0453593271
```

```
CV.With.Lass<- cv.gglasso(x = Las.x, y = Las.y, group = AspectGroup, pred.loss = "L1")
plot(CV.With.Lass)
```




```
coef(CV.With.Lass, s = "lambda.min")
```

```
##                        1
## (Intercept)  0.4623563860
## SoilTemp     0.0541986493
## SoilMoist    0.0074891926
## X            0.0011794055
## Y           -0.0005223101
## Z           -0.0085190466
## Slope        0.0013672451
## Aspect_E    -0.0140461170
## Aspect_N    -0.0024582480
## Aspect_NE   -0.0278372971
## Aspect_NW   -0.0190157892
## Aspect_S     0.0424722720
## Aspect_SE    0.0429493577
## Aspect_SW    0.0194214547
## Aspect_W    -0.0414852144
```

```
coef(CV.With.Lass, s = "lambda.1se")
```

```
##                     1
## (Intercept) 0.4536756
## SoilTemp    0.0000000
## SoilMoist   0.0000000
## X           0.0000000
## Y           0.0000000
## Z           0.0000000
## Slope       0.0000000
## Aspect_E    0.0000000
## Aspect_N    0.0000000
## Aspect_NE   0.0000000
## Aspect_NW   0.0000000
## Aspect_S    0.0000000
## Aspect_SE   0.0000000
## Aspect_SW   0.0000000
## Aspect_W    0.0000000
```

You won't see it here, but a problem with the cross validation is that it introduces some randomness. If you re run it, it comes to a different solution each time. I solve this a little ways down.

All Variables (Soil Data)

Okay, so, this is the second approach which prioritizes having all variables included at the expense of ~40 plots. Leaving us with a sample size of 64 instead of 107. But the soil data (pH, conductivity, and salinity) may be very important

Data Wrangling: Removing plots with NA for ph

The Lasso-ing process is much the same here as it was before. The tricky plot was really data wrangling. I'm sure there are a thousand better ways to wrangle but this works for me!

```
Las.x2<- With_Las %>%
  select(c( "SoilTemp", "SoilMoist","X", "Y", "Z", "Slope", "pH", "cond_us.cm", "salinity_pp
t")) %>% na.omit()


DumAspect2<-With_Las %>%
  dummy_cols( select_columns = "Aspect") %>%
  select(Aspect_E:Aspect_W, pH)%>%
  na.omit() %>%
  select(Aspect_E:Aspect_W)


Las.x3<- Las.x2%>%
  cbind(DumAspect2)

dim(Las.x3)
```

```
## [1] 64 17
```

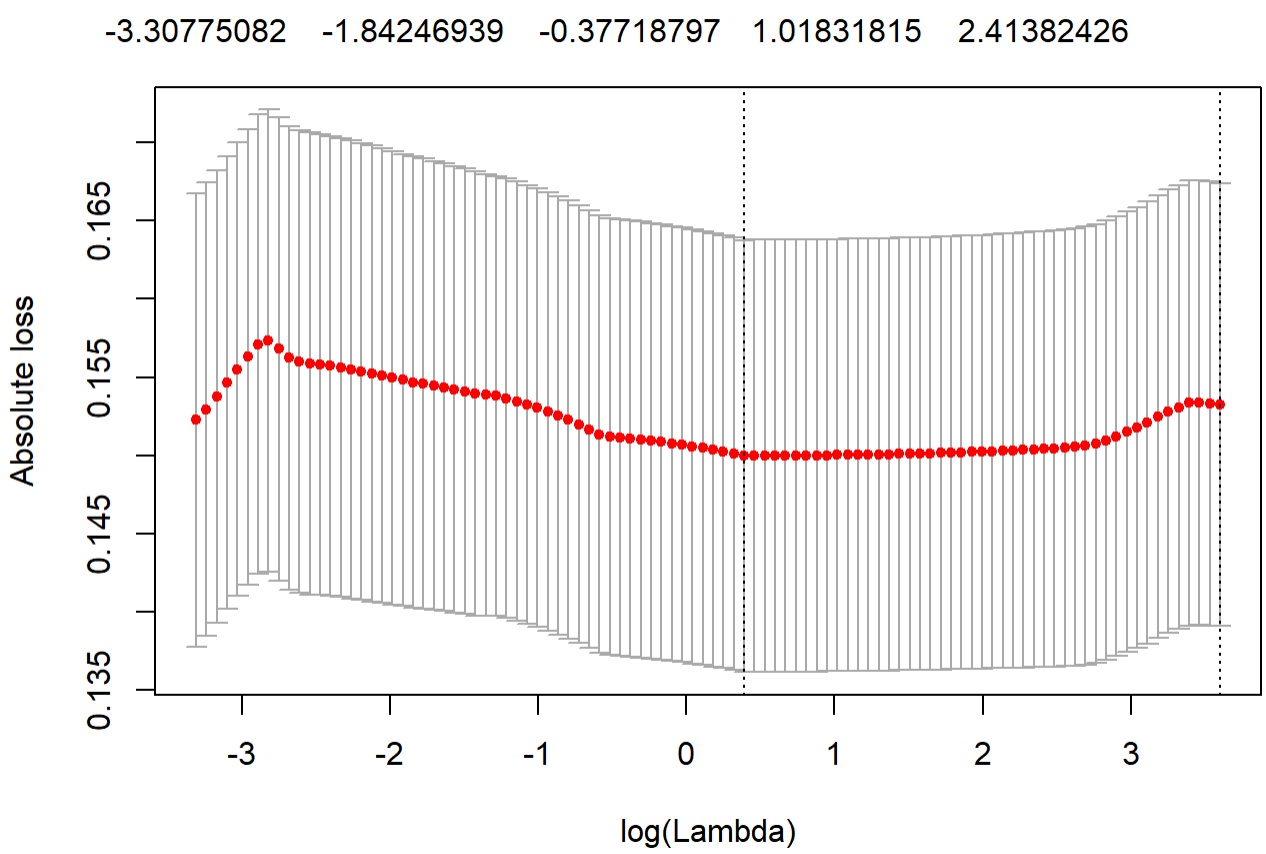
```
Las.y2 <- With_Las %>%
  select(Dissimilarity, pH) %>%
  na.omit() %>%
  select(Dissimilarity) %>%
  as.matrix()
dim(Las.y2)
```

```
## [1] 64  1
```

```
AspectGroup2 <- c((1:dim(Las.x2)[2]), rep(dim(Las.x2)[2]+1, 8))
dim(Las.x3)
```

```
## [1] 64 17
```

```
CV.With.Lass.Ph<- cv.gglasso(x = as.matrix(Las.x3), y = Las.y2, group = AspectGroup2, pred.lo
ss = "L1")
plot(CV.With.Lass.Ph)
```



```
coef(CV.With.Lass.Ph, s = "lambda.min")
```

```
##                          1
## (Intercept)   5.664465e-01
## SoilTemp      0.000000e+00
## SoilMoist     0.000000e+00
## X             0.000000e+00
## Y             0.000000e+00
## Z             0.000000e+00
## Slope         0.000000e+00
## pH            0.000000e+00
## cond_us.cm   -5.321168e-05
## salinity_ppt  0.000000e+00
## Aspect_E      0.000000e+00
## Aspect_N      0.000000e+00
## Aspect_NE     0.000000e+00
## Aspect_NW     0.000000e+00
## Aspect_S      0.000000e+00
## Aspect_SE     0.000000e+00
## Aspect_SW     0.000000e+00
## Aspect_W      0.000000e+00
```

```
coef(CV.With.Lass.Ph, s = "lambda.1se")
```

```
##                      1
## (Intercept)  0.5026862
## SoilTemp     0.0000000
## SoilMoist    0.0000000
## X            0.0000000
## Y            0.0000000
## Z            0.0000000
## Slope        0.0000000
## pH           0.0000000
## cond_us.cm   0.0000000
## salinity_ppt 0.0000000
## Aspect_E     0.0000000
## Aspect_N     0.0000000
## Aspect_NE    0.0000000
## Aspect_NW    0.0000000
## Aspect_S     0.0000000
## Aspect_SE    0.0000000
## Aspect_SW    0.0000000
## Aspect_W     0.0000000
```

Once again, we see that the 1se gives us zeros for everything so we are going proceed with the minimizing of lambda route. I've seen people argue for both, so whatever works!

Simulating

Okay, here is where the real magic starts. I noticed that there is some randomness in the cross-validated lasso which makes some sense. So, I figured, why not just run it a bunch of times and count up how often each variable was non-zero. So, that's what I did!

```
storage <- data.frame(row.names =  rownames(coef(CV.With.Lass, s = "lambda.1se")))
storage.pH<- data.frame(row.names = rownames(coef(CV.With.Lass.Ph, s = "lambda.1se")))
for(i in 1:10000){
  Loop <- cv.gglasso(x = Las.x, y = Las.y, group = AspectGroup, pred.loss = "L1")
  storage[,i] <-coef(Loop, s = "lambda.min")[,1]

  Loop.pH <- cv.gglasso(x = as.matrix(Las.x3), y = Las.y2, group = AspectGroup2, pred.loss =
"L1")
  storage.pH[,i] <-coef(Loop.pH, s = "lambda.min")[,1]

  i = i+1

}

rowSums(storage!=0)
```

```
## (Intercept)    SoilTemp   SoilMoist           X           Y           Z
##       10000        6012        2047        8897        9071        5015
##       Slope    Aspect_E    Aspect_N   Aspect_NE   Aspect_NW    Aspect_S
##        5556         583         583         583         583         583
##   Aspect_SE   Aspect_SW    Aspect_W
##         583         583         583
```

```
rowSums(storage.pH!=0)
```

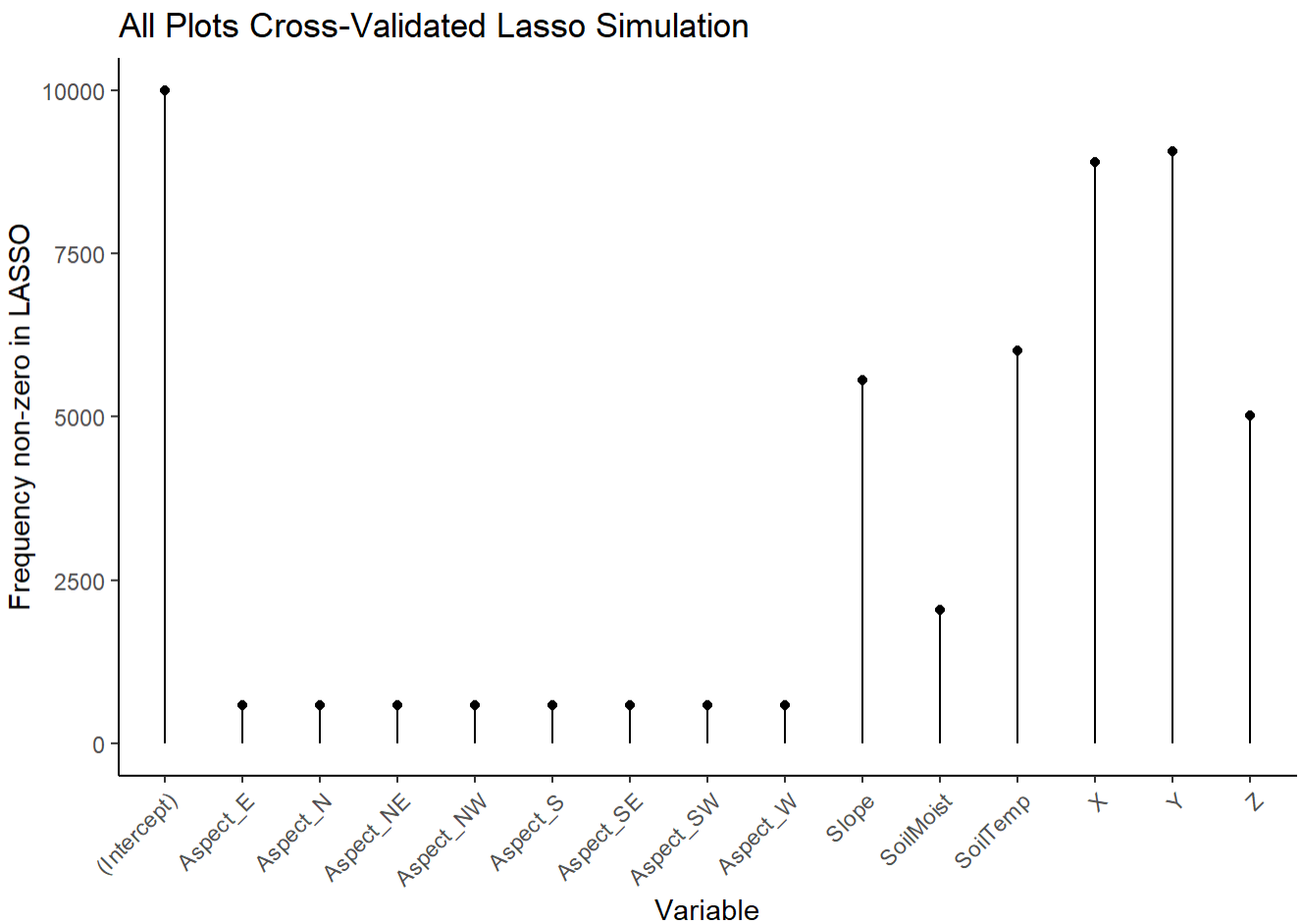
```
##  (Intercept)     SoilTemp    SoilMoist            X            Y            Z
##        10000         3342            0         3342         3518            0
##        Slope           pH   cond_us.cm salinity_ppt     Aspect_E     Aspect_N
##            0            0         8783            0            0            0
##    Aspect_NE    Aspect_NW     Aspect_S    Aspect_SE    Aspect_SW     Aspect_W
##            0            0            0            0            0            0
```

Okay! I've plotted and explained below!

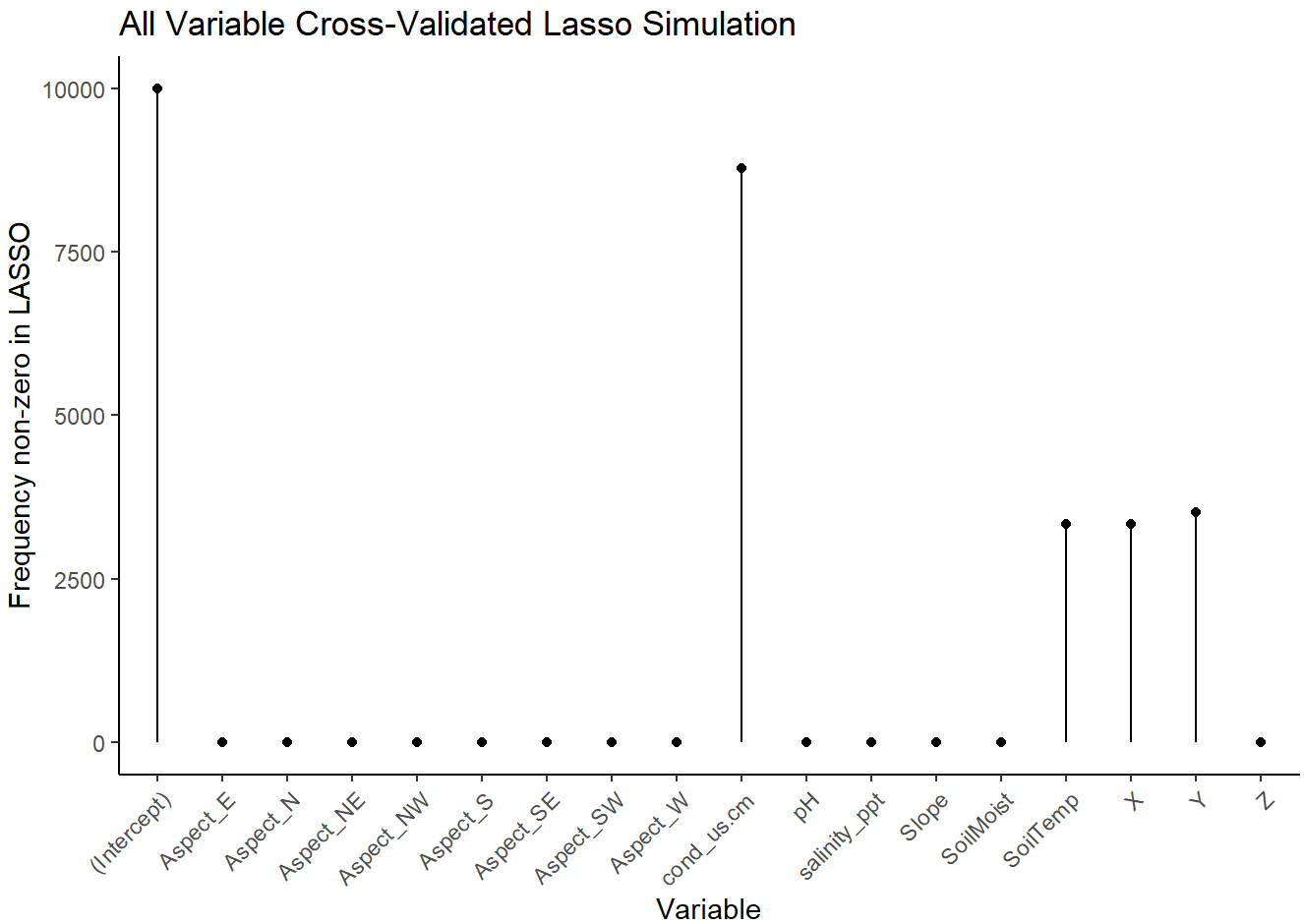
Plotting the Simulations

Just to make it easier to read

```
storage.plot<- data.frame(Name = rownames_to_column(storage)[1], totals = rowSums(storage!=
0))
storage.pH.plot<- data.frame(Name = rownames_to_column(storage.pH)[1], totals = rowSums(stora
ge.pH!=0))


ggplot(aes(x = rowname, y = totals), data = storage.plot)+
  geom_point()+
  geom_segment( aes(x=rowname, xend=rowname, y=0, yend=totals))+
  scale_x_discrete(guide = guide_axis(angle = 45)) +
  theme_classic()+
  labs(x = "Variable", y = "Frequency non-zero in LASSO", title = "All Plots Cross-Validated
Lasso Simulation")
```




```
ggplot(aes(x = rowname, y = totals), data = storage.pH.plot)+
  geom_point()+
  geom_segment( aes(x=rowname, xend=rowname, y=0, yend=totals))+
  scale_x_discrete(guide = guide_axis(angle = 45)) +
  theme_classic()+
  labs(x = "Variable", y = "Frequency non-zero in LASSO", title = "All Variable Cross-Validat
ed Lasso Simulation")
```



This is where everything clicked into place for me! Having the all variable simulation spit out only four variables that matter helped guide me in the all plots variant. Starting with the four most common variable in the all plots area was great guidance!

Using LASSO

Now that I have my starting points! I can do more classic model selection where I can really justify what I'm doing and why.

All Plots

The simulation for the All Plots direction spit our the Y axis (which is weird bc it shouldn't matter), the X axis (distance from waterline), Soil Temperature (it simply had to), and Slope (Okay) as the variables that most frequently mattered. That's a great place to start!

Model Selection

So, here's the deal. I have known that Soil Temperature is the only thing that matter, but I haven't been able to justify what model to put it in and how I got there. A full model was being beaten by a null model and that wasn't right. Here, I put all four in a model then compared it to a null model and then each of the four individually and none of them beat them (some tied, but I think I can justify keeping the full). I then did the same negligible effects plot and everything as before!

```
Lasso.plots.full <- lmer(Dissimilarity ~ Y + X + SoilTemp + Slope + (1|Site), data = With_La
s)

Lasso.plots.null <- lmer(Dissimilarity ~ 1  + (1|Site), data = With_Las)
Lasso.plots.ST<- lmer(Dissimilarity ~ SoilTemp  + (1|Site), data = With_Las)
Lasso.plots.Y<- lmer(Dissimilarity ~ Y  + (1|Site), data = With_Las)
Lasso.plots.X<- lmer(Dissimilarity ~ X  + (1|Site), data = With_Las)
Lasso.plots.Slope<- lmer(Dissimilarity ~ Slope  + (1|Site), data = With_Las)

summary(Lasso.plots.full)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Dissimilarity ~ Y + X + SoilTemp + Slope + (1 | Site)
##    Data: With_Las
##
## REML criterion at convergence: -34.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.9535 -0.6590 -0.1467  0.5295  3.3249
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  Site     (Intercept) 0.005343 0.0731
##  Residual             0.026750 0.1636
## Number of obs: 107, groups:  Site, 3
##
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept)  0.4600607  0.0658936   6.982
## Y           -0.0003826  0.0003260  -1.173
## X            0.0006793  0.0007484   0.908
## SoilTemp     0.0588831  0.0168569   3.493
## Slope        0.0005367  0.0017622   0.305
##
## Correlation of Fixed Effects:
##          (Intr) Y      X      SolTmp
## Y        -0.377
## X        -0.497  0.009
## SoilTemp -0.030 -0.014 -0.073
## Slope    -0.360 -0.091  0.030  0.246
```

```
anova(Lasso.plots.full, Lasso.plots.null)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.null: Dissimilarity ~ 1 + (1 | Site)
## Lasso.plots.full: Dissimilarity ~ Y + X + SoilTemp + Slope + (1 | Site)
##                  npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.plots.null    3 -62.006 -53.987 34.003  -68.006
## Lasso.plots.full    7 -68.865 -50.155 41.432  -82.865 14.859  4   0.005003 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(Lasso.plots.full, Lasso.plots.ST) #technically has a lower AIC, but not sig
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.ST: Dissimilarity ~ SoilTemp + (1 | Site)
## Lasso.plots.full: Dissimilarity ~ Y + X + SoilTemp + Slope + (1 | Site)
##                  npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.plots.ST      4 -72.478 -61.786 40.239  -80.478
## Lasso.plots.full    7 -68.865 -50.155 41.432  -82.865 2.3869  3     0.4961
```

```
anova(Lasso.plots.full, Lasso.plots.Y)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.Y: Dissimilarity ~ Y + (1 | Site)
## Lasso.plots.full: Dissimilarity ~ Y + X + SoilTemp + Slope + (1 | Site)
##                  npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.plots.Y       4 -61.329 -50.637 34.664  -69.329
## Lasso.plots.full    7 -68.865 -50.155 41.432  -82.865 13.536  3    0.00361 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(Lasso.plots.full, Lasso.plots.X)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.X: Dissimilarity ~ X + (1 | Site)
## Lasso.plots.full: Dissimilarity ~ Y + X + SoilTemp + Slope + (1 | Site)
##                  npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.plots.X       4 -61.416 -50.725 34.708  -69.416
## Lasso.plots.full    7 -68.865 -50.155 41.432  -82.865 13.449  3    0.00376 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(Lasso.plots.full, Lasso.plots.Slope)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.Slope: Dissimilarity ~ Slope + (1 | Site)
## Lasso.plots.full: Dissimilarity ~ Y + X + SoilTemp + Slope + (1 | Site)
##                   npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.plots.Slope    4 -60.443 -49.752 34.222  -68.443
## Lasso.plots.full     7 -68.865 -50.155 41.432  -82.865 14.421  3   0.002384 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

THE FULL MODEL WINS HELL FUCKING YES

Technically the Soil Temperature alone model has a lower AIC, but not significantly so!

Another Attempt at model Selection

I had another idea for how to do data selection by adding a variable from most to least common. Same result!

Okay, I tried another model selection approach

Since there was more variablity in the Lasso simulation for the All Plots path, I decided to do another style of model selection where I started with the most common and then added up. It gives the same result so you can skip this section if you want.

```
Lasso.plots.null <- lmer(Dissimilarity ~ 1  + (1|Site), data = With_Las)

Lasso.plots.Y<- lmer(Dissimilarity ~ Y  + (1|Site), data = With_Las)

anova(Lasso.plots.Y, Lasso.plots.null)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.null: Dissimilarity ~ 1 + (1 | Site)
## Lasso.plots.Y: Dissimilarity ~ Y + (1 | Site)
##                  npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.plots.null    3 -62.006 -53.987 34.003  -68.006
## Lasso.plots.Y       4 -61.329 -50.637 34.664  -69.329 1.3226  1     0.2501
```

```
anova(Lasso.plots.X, Lasso.plots.null)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.null: Dissimilarity ~ 1 + (1 | Site)
## Lasso.plots.X: Dissimilarity ~ X + (1 | Site)
##                  npar     AIC     BIC logLik deviance Chisq Df Pr(>Chisq)
## Lasso.plots.null    3 -62.006 -53.987 34.003  -68.006
## Lasso.plots.X       4 -61.416 -50.725 34.708  -69.416  1.41  1     0.2351
```

```
Lasso.plots.YXst<- lmer(Dissimilarity ~ Y + X + SoilTemp + (1|Site), data = With_Las)


anova(Lasso.plots.YXst, Lasso.plots.null)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.null: Dissimilarity ~ 1 + (1 | Site)
## Lasso.plots.YXst: Dissimilarity ~ Y + X + SoilTemp + (1 | Site)
##                  npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.plots.null    3 -62.006 -53.987 34.003  -68.006
## Lasso.plots.YXst    6 -70.749 -54.712 41.374  -82.749 14.743  3    0.00205 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Lasso.plots.YXstSlopeZ<- lmer(Dissimilarity ~ Y + X + SoilTemp + Slope+ Z+ (1|Site), data = W
ith_Las)
anova(Lasso.plots.YXst, Lasso.plots.YXstSlopeZ)
```

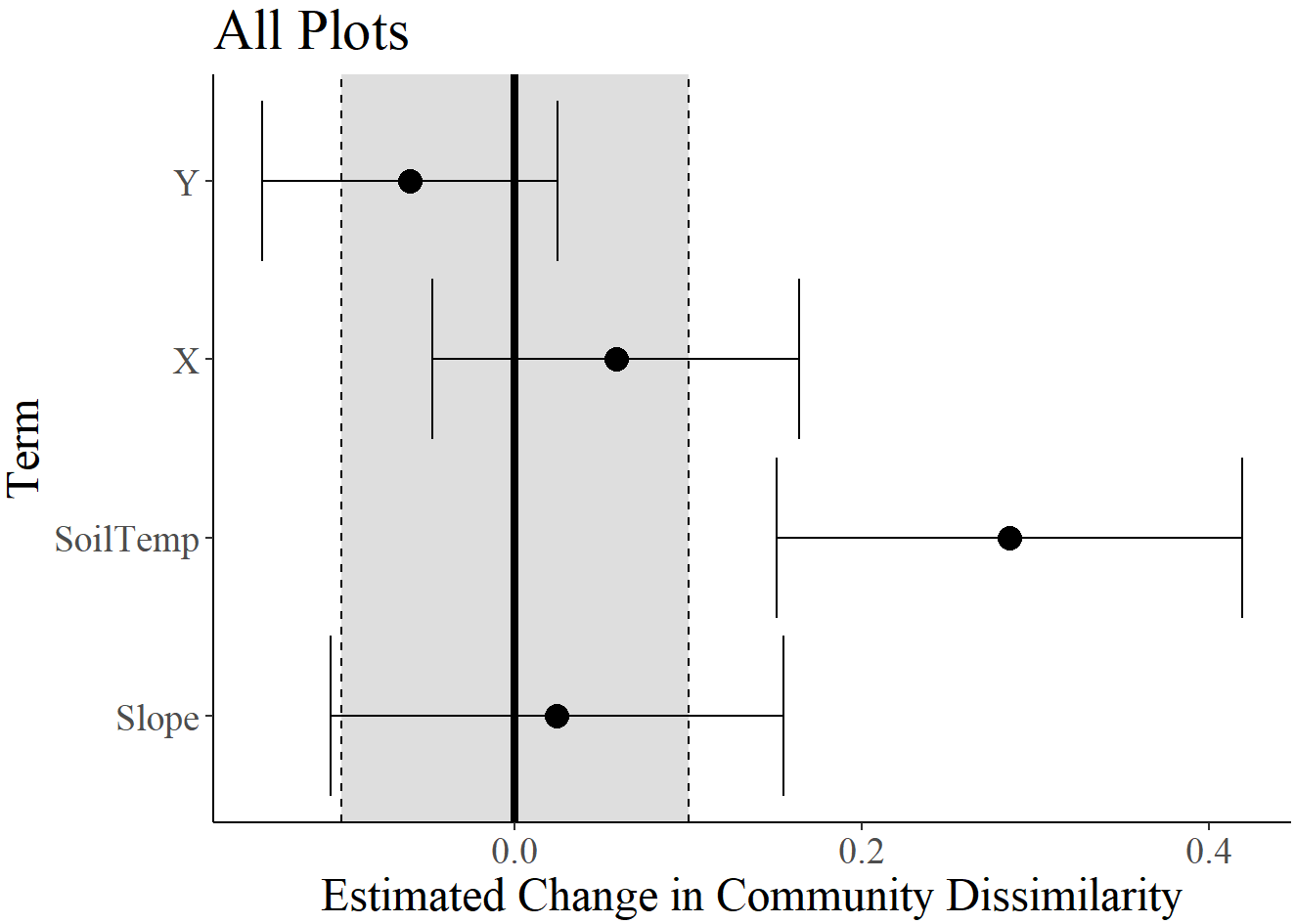
```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las
## Models:
## Lasso.plots.YXst: Dissimilarity ~ Y + X + SoilTemp + (1 | Site)
## Lasso.plots.YXstSlopeZ: Dissimilarity ~ Y + X + SoilTemp + Slope + Z + (1 | Site)
##                        npar     AIC     BIC logLik deviance  Chisq Df
## Lasso.plots.YXst          6 -70.749 -54.712 41.374  -82.749
## Lasso.plots.YXstSlopeZ    8 -67.936 -46.553 41.968  -83.936 1.1869  2
##                        Pr(>Chisq)
## Lasso.plots.YXst
## Lasso.plots.YXstSlopeZ     0.5524
```

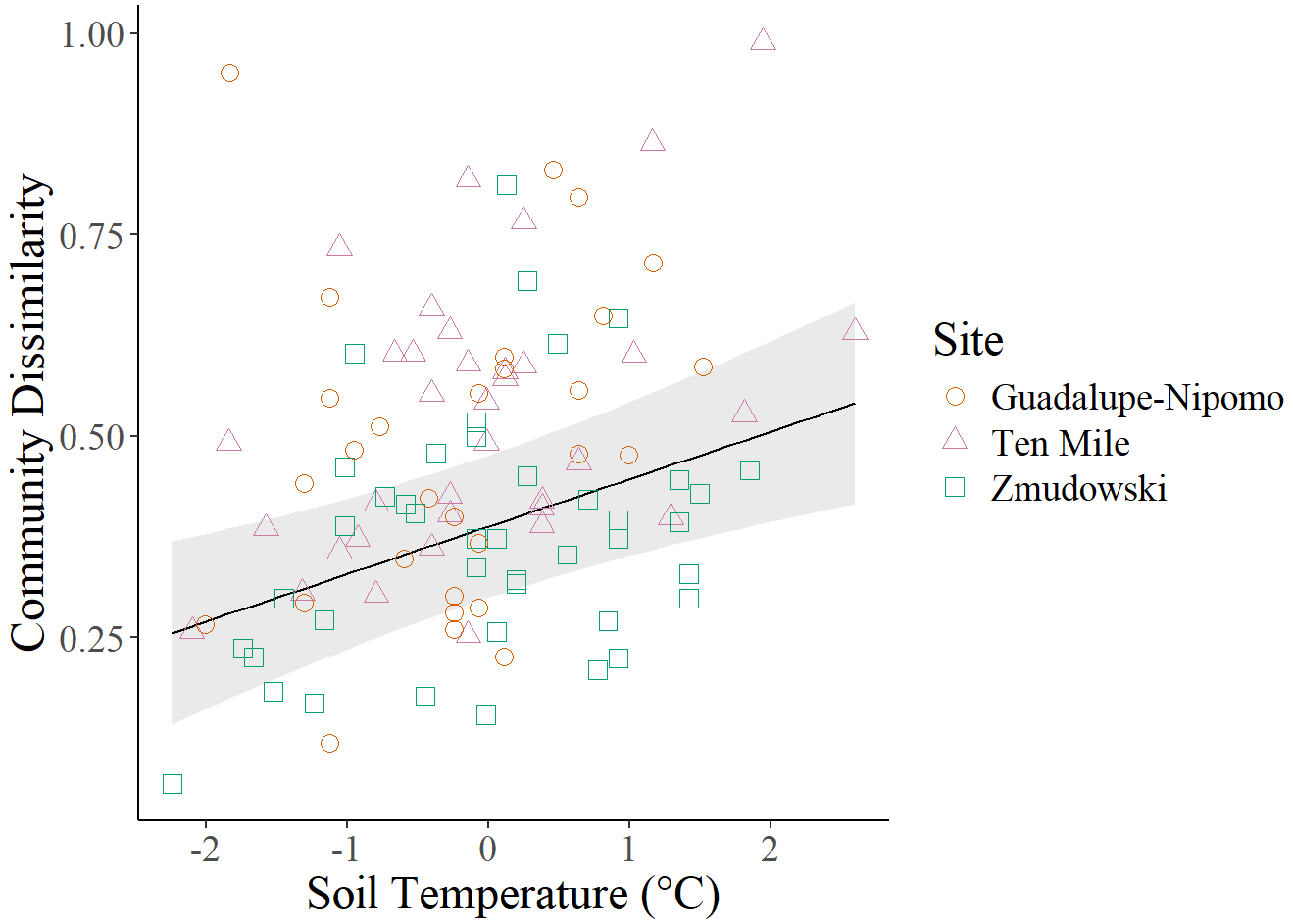
Negligbile effects

Same process as before and same story 😍

```
Lasso.cmp<- avg_comparisons(Lasso.plots.full, variables = list(Y = "minmax", X = "minmax", So
ilTemp = "minmax", Slope = "minmax"),   conf_level = 0.90, type = "response")

Lasso.cmp <- as.data.frame(Lasso.cmp)

ggplot(Lasso.cmp, aes(x = estimate,
                 xmin = conf.low,
                 xmax = conf.high,
                 y = term,
                 group = contrast)) +
     annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
  geom_vline(xintercept = c(-0.1, .1), linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "solid", linewidth = 1.5)+
  geom_errorbarh(aes(xmin = conf.low,  xmax = conf.high,)) +
  geom_point(size = 4) +
  theme_classic()+
  labs(x = "Estimated Change in Community Dissimilarity", y = "Term", title = "All Plots")+
  theme(text = element_text(family = "serif", size = 18))
```




```
plot_predictions(Lasso.plots.full, condition = "SoilTemp")+
  geom_point(data = subset(Comb_Dis, Dissimilarity >0), size = 3,aes(x = SoilTemp, y = Dissim
ilarity, color = Site, shape = Site))+
  labs(x = "Soil Temperature (°C)", y =  "Community Dissimilarity")+
  scale_color_manual(labels = c("Guadalupe-Nipomo", "Ten Mile", "Zmudowski"), values = c("#d5
5e00", "#cc79a7", "#009e73"), name = "Site")+
  scale_shape_manual(labels = c("Guadalupe-Nipomo", "Ten Mile", "Zmudowski"), values = c
(1,2,0), name = "Site")+
  theme_classic()+
  theme(text = element_text(family = "serif", size = 18))
```



All Variables

We repeat the same process for the All Variables pathway. This was much easier because there were only four variables that ever mattered: Soil conductivity, Y axis, X axis, and Soil Temperature. We start there and then see if it is any good.

Model Selection

```
With_Las_Soil <- subset(With_Las, !is.na(cond_us.cm))

Lasso.vars.full <- lmer(Dissimilarity ~ cond_us.cm + Y + X + SoilTemp + (1|Site), data = With
_Las_Soil)

Lasso.vars.null <- lmer(Dissimilarity ~ 1  + (1|Site), data = With_Las_Soil)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
Lasso.vars.ST<- lmer(Dissimilarity ~ SoilTemp  + (1|Site), data = With_Las_Soil)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
Lasso.vars.Y<- lmer(Dissimilarity ~ Y  + (1|Site), data = With_Las_Soil)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
Lasso.vars.X<- lmer(Dissimilarity ~ X  + (1|Site), data = With_Las_Soil)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
Lasso.vars.Cond<- lmer(Dissimilarity ~ cond_us.cm  + (1|Site), data = With_Las_Soil)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(Lasso.vars.full)
```

```
## Linear mixed model fit by REML ['lmerMod']
## Formula: Dissimilarity ~ cond_us.cm + Y + X + SoilTemp + (1 | Site)
##    Data: With_Las_Soil
##
## REML criterion at convergence: 6.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -1.5635 -0.6374 -0.1791  0.4860  2.9935
##
## Random effects:
##  Groups   Name        Variance  Std.Dev.
##  Site     (Intercept) 0.0002717 0.01648
##  Residual             0.0288462 0.16984
## Number of obs: 64, groups:  Site, 2
##
## Fixed effects:
##               Estimate Std. Error t value
## (Intercept)  6.133e-01  7.457e-02   8.226
## cond_us.cm  -5.224e-05  2.863e-05  -1.825
## Y           -4.443e-04  4.565e-04  -0.973
## X           -1.986e-05  1.069e-03  -0.019
## SoilTemp     7.202e-02  2.255e-02   3.194
##
## Correlation of Fixed Effects:
##            (Intr) cnd_s. Y      X
## cond_us.cm -0.501
## Y          -0.445 -0.148
## X          -0.706  0.178  0.016
## SoilTemp    0.041  0.061 -0.167  0.095
```

```
anova(Lasso.vars.full, Lasso.vars.null)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las_Soil
## Models:
## Lasso.vars.null: Dissimilarity ~ 1 + (1 | Site)
## Lasso.vars.full: Dissimilarity ~ cond_us.cm + Y + X + SoilTemp + (1 | Site)
##                 npar     AIC     BIC logLik deviance Chisq Df Pr(>Chisq)
## Lasso.vars.null    3 -29.949 -23.473 17.975  -35.949
## Lasso.vars.full    7 -36.359 -21.247 25.180  -50.359 14.41  4   0.006095 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(Lasso.vars.full, Lasso.vars.ST)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las_Soil
## Models:
## Lasso.vars.ST: Dissimilarity ~ SoilTemp + (1 | Site)
## Lasso.vars.full: Dissimilarity ~ cond_us.cm + Y + X + SoilTemp + (1 | Site)
##                 npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.vars.ST      4 -37.140 -28.504  22.57  -45.140
## Lasso.vars.full    7 -36.359 -21.247  25.18  -50.359 5.2197  3     0.1564
```

```
anova(Lasso.vars.full, Lasso.vars.Y)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las_Soil
## Models:
## Lasso.vars.Y: Dissimilarity ~ Y + (1 | Site)
## Lasso.vars.full: Dissimilarity ~ cond_us.cm + Y + X + SoilTemp + (1 | Site)
##                 npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.vars.Y       4 -28.457 -19.821 18.229  -36.457
## Lasso.vars.full    7 -36.359 -21.247 25.180  -50.359 13.902  3   0.003041 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(Lasso.vars.full, Lasso.vars.X)
```

```
## refitting model(s) with ML (instead of REML)
```

```
## Data: With_Las_Soil
## Models:
## Lasso.vars.X: Dissimilarity ~ X + (1 | Site)
## Lasso.vars.full: Dissimilarity ~ cond_us.cm + Y + X + SoilTemp + (1 | Site)
##                 npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.vars.X       4 -27.956 -19.320 17.978  -35.956
## Lasso.vars.full    7 -36.359 -21.247 25.180  -50.359 14.404  3   0.002404 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

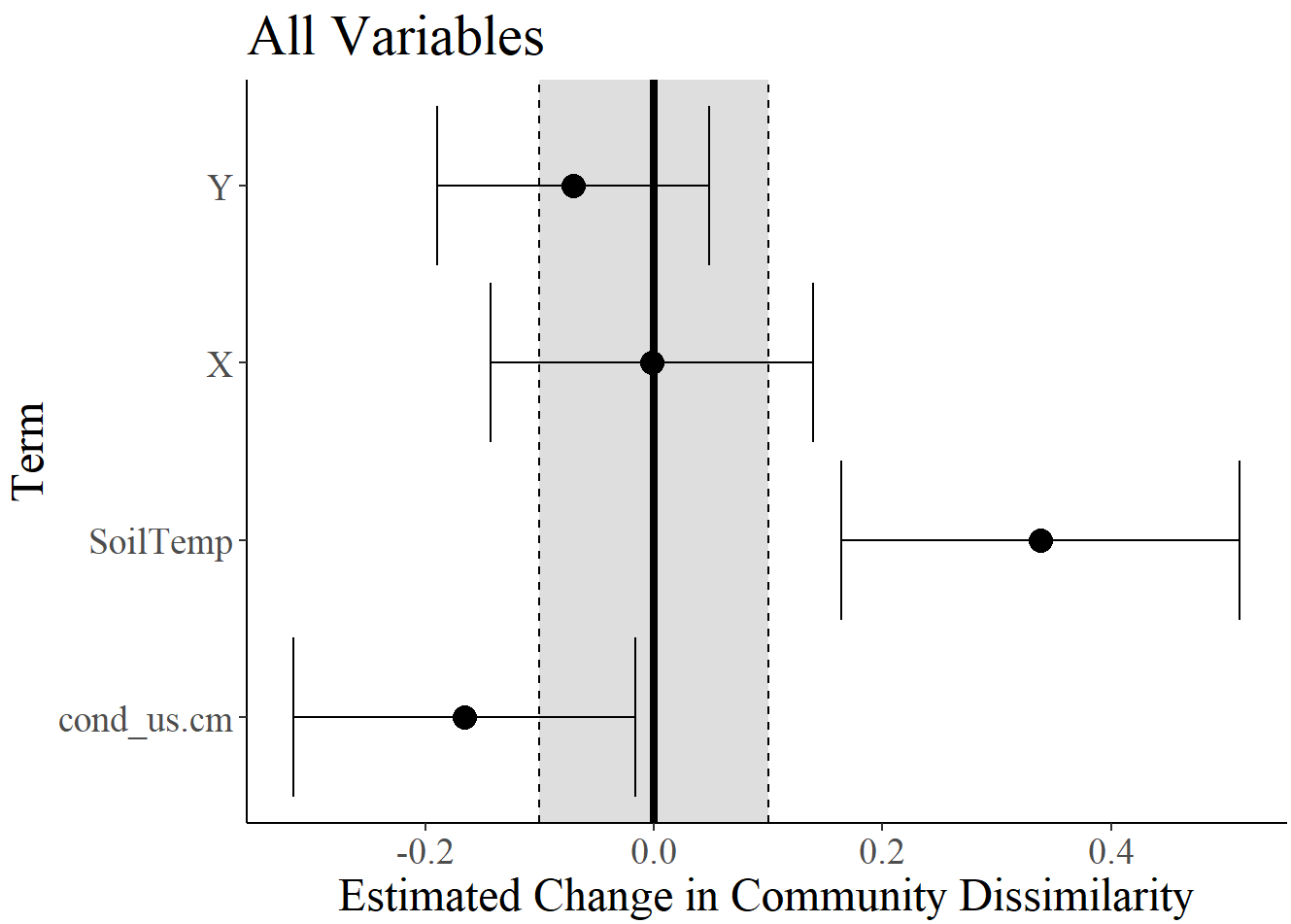
```
anova(Lasso.vars.full, Lasso.vars.Cond)
```

```
## refitting model(s) with ML (instead of REML)
```

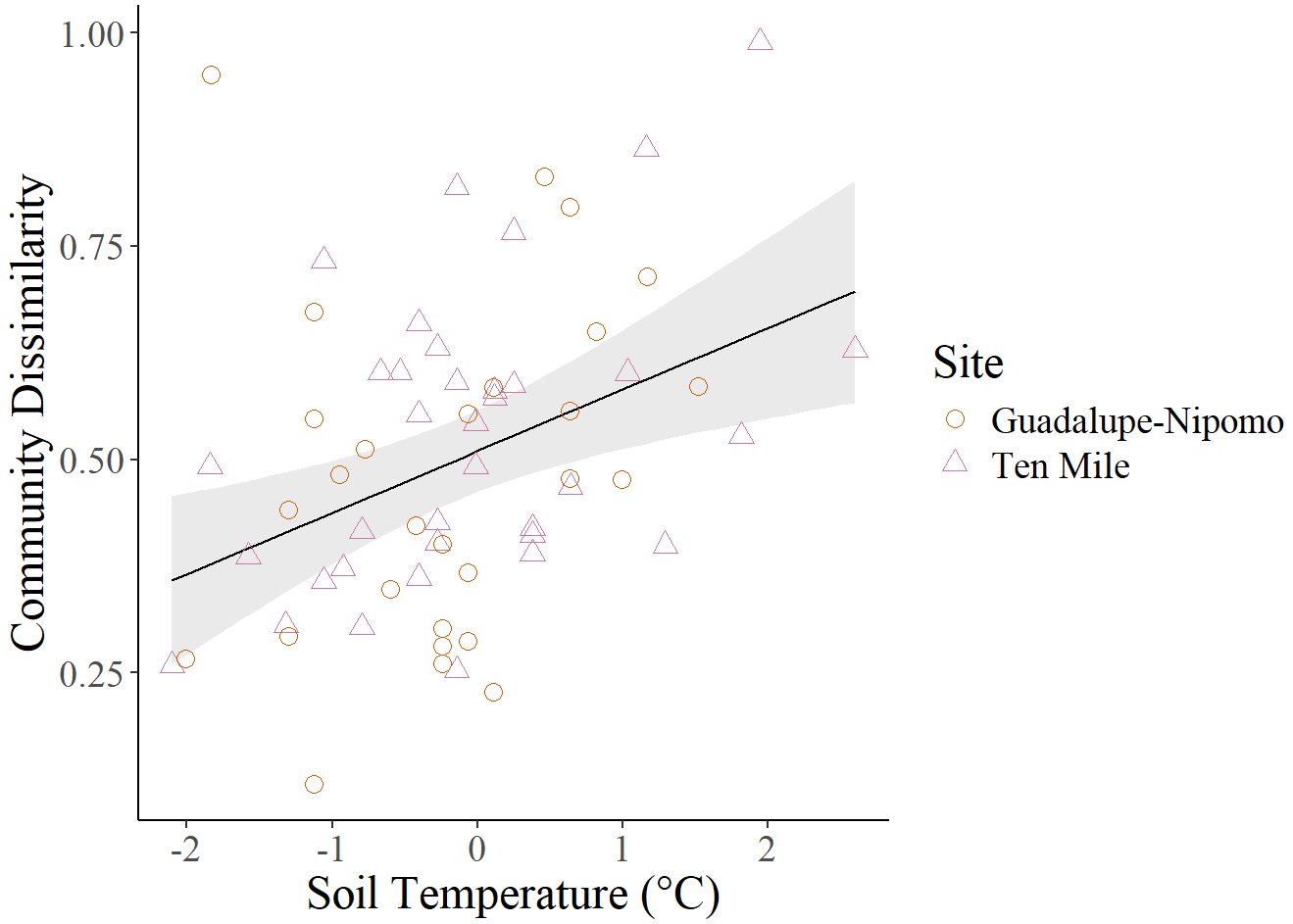
```
## Data: With_Las_Soil
## Models:
## Lasso.vars.Cond: Dissimilarity ~ cond_us.cm + (1 | Site)
## Lasso.vars.full: Dissimilarity ~ cond_us.cm + Y + X + SoilTemp + (1 | Site)
##                 npar     AIC     BIC logLik deviance  Chisq Df Pr(>Chisq)
## Lasso.vars.Cond    4 -31.963 -23.327 19.981  -39.963
## Lasso.vars.full    7 -36.359 -21.247 25.180  -50.359 10.397  3    0.01548 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Same story with Soil Temperature but yay full model!!

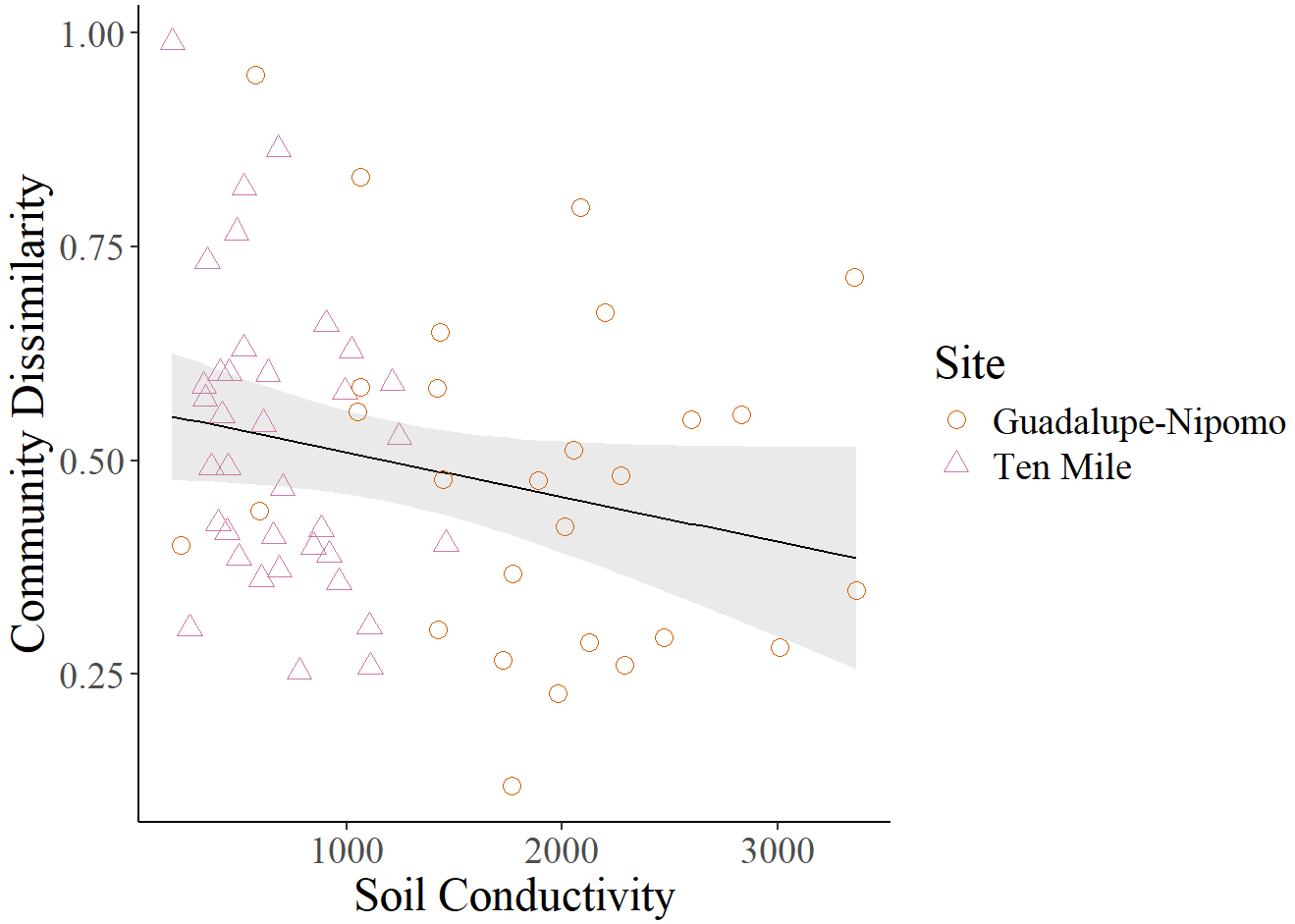
```
Lasso.vars.cmp<- avg_comparisons(Lasso.vars.full, variables = list(Y = "minmax", X = "minma
x", SoilTemp = "minmax", cond_us.cm = "minmax"),   conf_level = 0.90, type = "response")

Lasso.vars.cmp <- as.data.frame(Lasso.vars.cmp)

ggplot(Lasso.vars.cmp, aes(x = estimate,
                xmin = conf.low,
                xmax = conf.high,
                y = term,
                group = contrast)) +
     annotate("rect", xmin = -0.1, xmax = 0.1, ymin = -Inf, ymax = Inf, alpha = .2)+
  geom_vline(xintercept = c(-0.1, .1), linetype = "dashed") +
  geom_vline(xintercept = 0, linetype = "solid", linewidth = 1.5)+
  geom_errorbarh(aes(xmin = conf.low,  xmax = conf.high,)) +
  geom_point(size = 4) +
  theme_classic()+
  labs(x = "Estimated Change in Community Dissimilarity", y = "Term", title = "All Variable
s")+
  theme(text = element_text(family = "serif", size = 18))
```




```
plot_predictions(Lasso.vars.full, condition = "SoilTemp")+
  geom_point(data = With_Las_Soil, size = 3,aes(x = SoilTemp, y = Dissimilarity, color = Site
, shape = Site))+
  labs(x = "Soil Temperature (°C)", y =  "Community Dissimilarity")+
  scale_color_manual(labels = c("Guadalupe-Nipomo", "Ten Mile", "Zmudowski"), values = c("#d5
5e00", "#cc79a7", "#009e73"), name = "Site")+
  scale_shape_manual(labels = c("Guadalupe-Nipomo", "Ten Mile", "Zmudowski"), values = c
(1,2,0), name = "Site")+
  theme_classic()+
  theme(text = element_text(family = "serif", size = 18))
```




```
plot_predictions(Lasso.vars.full, condition = "cond_us.cm")+
  geom_point(data = With_Las_Soil, size = 3,aes(x = cond_us.cm, y = Dissimilarity, color = Si
te, shape = Site))+
  labs(x = "Soil Conductivity", y =  "Community Dissimilarity")+
  scale_color_manual(labels = c("Guadalupe-Nipomo", "Ten Mile", "Zmudowski"), values = c("#d5
5e00", "#cc79a7", "#009e73"), name = "Site")+
  scale_shape_manual(labels = c("Guadalupe-Nipomo", "Ten Mile", "Zmudowski"), values = c
(1,2,0), name = "Site")+
  theme_classic()+
  theme(text = element_text(family = "serif", size = 18))
```



INTERESTING!!!! Soil conductivity is signifgant but unsure if it is substantial. But certainly worth reporting and discussing! New results! This one actually conforms to the Stress Gradient Hypothesis better so the story gets nice and muddy!

Conculsion

This feels like a really justifiable and useful and interesing way to present these data at this level! Complicates things a smidge, but I think it is really worth it! However, let me know if there is something I'm not seeing here, but I'm pretty confident in defending it. Super cool and thank you for the kick in the pants!

-B